This story illustrates a number of important points which came up in the previous chapter. It shows that it can be perfectly proper to speak of a 'gene for behaviour so-and-so' even if we haven't the faintest idea of the chemical chain of embryonic causes leading from gene to behaviour. The chain of causes could even turn out to involve learning. For example, it could be that the uncapping gene exerts its effect by giving bees a taste for infected wax. This means they will find the eating of the wax caps covering disease-victims rewarding, and will therefore tend to repeat it. Even if this is how the gene works, it is still truly a gene 'for uncapping' provided that, other things being equal, bees possessing the gene end up by uncapping, and bees not possessing the gene do not uncap.

Secondly it illustrates the fact that genes 'cooperate' in their effects on the behaviour of the communal survival machine. The throwing-out gene is useless unless it is accompanied by the uncapping gene and vice versa. Yet the genetic experiments show equally clearly that the two genes are in principle quite separable in their journey through the generations. As far as their useful work is concerned you can think of them as a single cooperating unit, but as replicating genes they are two free and independent agents.

For purposes of argument it will be necessary to speculate about genes 'for' doing all sorts of improbable things. If I speak, for example, of a hypothetical gene 'for saving companions from drowning', and you find such a concept incredible, remember the story of the hygienic bees. Recall that we are not talking about the gene as the sole antecedent cause of all the complex muscular contractions, sensory integrations, and even conscious decisions that are involved in saving somebody from drowning. We are saying nothing about the question of whether learning, experience, or environmental influences enter into the development of the behaviour. All you have to concede is that it is possible for a single gene, other things being equal and lots of other essential genes and environmental factors being present, to make a body more likely to save somebody from drowning than its allele would. The difference between the two genes may turn out at bottom to be a slight difference in some simple quantitative variable. The details of the embryonic developmental process, interesting as they may be, are irrelevant to evolutionary considerations. Konrad Lorenz has put this point well.

The genes are master programmers, and they are programming for their lives. They are judged according to the success of their programs in